# The Control Is the Instrument

Calibrating Perturbation Tests Used to Prove That EEG Decoders Read the Brain

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A standard argument in brain–computer-interface research runs: we removed the $\mu$ band, accuracy dropped, therefore the decoder uses sensorimotor rhythm. The argument is valid only if the perturbation has known sensitivity and, critically, a known false-alarm rate for the decoder being tested. Almost no published perturbation control has been characterised that way. We treat controls as measuring instruments and calibrate them against operational ground truth. Using a simulator in which the signal is planted by construction, we measure each control's catch rate and false-alarm rate across a grid of conditions and four decoder families. The reliability of a control turns out to depend on the decoder, not just on the control. Averaged over probes, EEGNet's false-alarm rate is 1.3% while band-power's is 68.8%: on the same simulated data, with the same perturbations, the identical evidence is decisive for one decoder and near-worthless for another. A random forest reading exactly the same features as a band-power linear model cuts spatial false alarms roughly in half while remaining equally fooled by band perturbations, showing the failure is jointly a property of the feature representation and the model class. Applying the calibrated battery to real PhysioNet decoders yields opposite trust verdicts for EEGNet and band-power on identical data. Finally, we grade the controls on real brains using eyes-open versus eyes-closed, a task whose occipital-alpha ground truth is fixed by physiology: band-power false-alarms on three of four controls whose targets are absent and CSP on two, exactly as the simulator predicted.

CCS Concepts: • **Applied computing** → **Life and medical sciences**; • **Computing methodologies** → *Machine learning*; *Modeling and simulation*.

Additional Key Words and Phrases: brain–computer interfaces, EEG, explainability, perturbation tests, simulation, validation

## 1 INTRODUCTION

Decoders that read motor imagery from scalp EEG are routinely defended with perturbation evidence. The analyst removes a frequency band, scrambles or silences a set of electrodes, and reports that accuracy fell; the fall is then offered as proof that the decoder depends on genuine neural activity in that band or region. The logic is appealing because it appears causal—an intervention was performed and an effect was observed.

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The logic is incomplete. A perturbation test is a diagnostic instrument, and an instrument that has never been characterised has no known error rate. To interpret "the control fired" one needs two numbers: the probability that it fires when the targeted dependence is genuinely present (sensitivity), and the probability that it fires when the targeted dependence is genuinely absent (false-alarm rate). Without the second number, a positive result is uninterpretable, because a control that fires under any perturbation whatsoever carries no information about which perturbation was applied.

Measuring those two numbers requires knowing the truth, and on real brains the truth is exactly what is unknown. This is the impasse we resolve by simulation.

Our concern parallels a line of work in machine-learning interpretability showing that attribution methods can pass visual inspection while failing basic sanity checks [1, 7], that occlusion-style evaluations need careful controls [6], and that model weights in neuroimaging admit no direct interpretation as signal sources [5, 13]. Perturbation controls in BCI have not received the same scrutiny.

### 1.1 Contributions

(1) An instrument-calibration framing of perturbation controls, in which a control is characterised by a measured (sensitivity, false-alarm) pair rather than assumed valid because it is intuitive.
(2) A controllable MI-EEG simulator with operational ground truth, and measured operating characteristics for band and spatial controls across four decoder families.
(3) The central empirical finding that control reliability is *decoder-conditional*, with a factor of 50 separating the best and worst false-alarm rates on identical data, plus a controlled dissociation isolating how much of the failure is due to features and how much to model class.
(4) A calibrated audit of real PhysioNet decoders that reaches opposite trust verdicts for EEGNet and band-power on the same recordings.
(5) A real-brain validation on eyes-open/eyes-closed, a task whose ground truth is fixed by physiology, confirming the simulator's specificity warning outside simulation.

## 2 A SIMULATOR WITH OPERATIONAL GROUND TRUTH

### 2.1 Generative model

Each trial is a matrix $X \in \mathbb{R}^{C\times T}$ with $C = 19$ electrodes at $F_s = 128$ Hz over $T = 256$ samples (2 s). The background is per-channel pink noise, matching the $1/f$ character of real EEG [14]: with $\tilde{n}_c(f)$ complex Gaussian,

$$n_c(t) = \mathcal{F}^{-1}\left\{\tilde{n}_c(f)\, f^{-1/2}\right\}(t), \qquad n_c \leftarrow n_c/\mathrm{std}(n_c). \tag{1}$$

The planted signal is a windowed 10 Hz $\mu$ burst added only to the motor channels contralateral to the imagined hand,

$$X_{c,t} = n_{c,t} + s\,\mathbb{1}[c \in \mathcal{M}(y)]\;w(t)\sin\left(2\pi f_\mu t/F_s\right), \tag{2}$$

where $w$ is a Hann window, $f_\mu = 10$ Hz, $\mathcal{M}(1)$ is the left-motor set and $\mathcal{M}(0)$ the right-motor set, and $s$ is the signal-strength knob. Setting $s = 0$ produces data with *provably* no class-discriminative signal anywhere, which is the regime in which false alarms are defined. Equation (2) reproduces the lateralised event-related desynchronisation that motor-imagery decoders exploit [11].

Figure 1 shows the testbed: the montage, the spectral effect of the planted burst, and the resulting ground-truth topography, which sits on the motor channels and nowhere else.

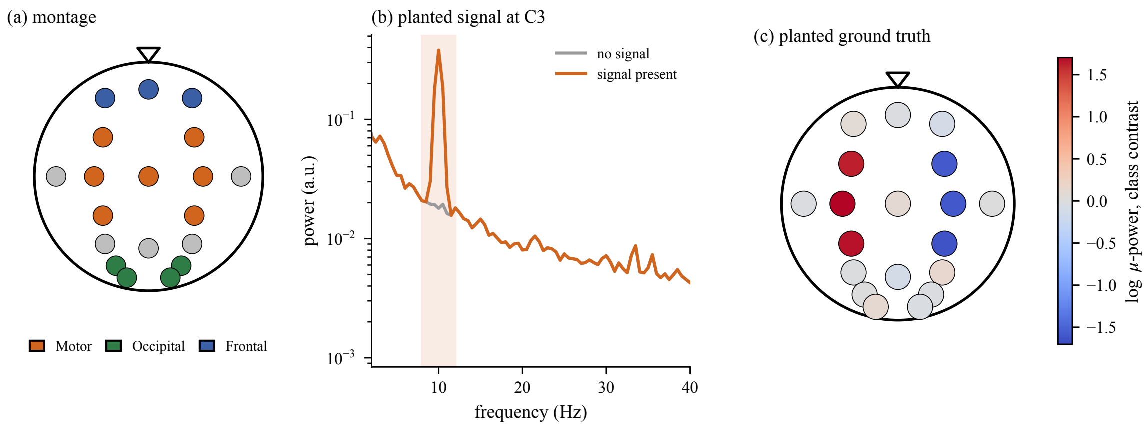


Fig. 1. The simulator. (a) The 19-electrode montage with region labels. (b) Welch spectra at C3 with and without the planted signal; the $\mu$-band bump is present only when $s > 0$. (c) The planted ground truth: the class contrast in $\mu$ power is confined to motor channels by construction.

### 2.2 Ground truth is operational, not biological

We are explicit that Equation (2) is not a cortex. Its value is *identifiability*: because we placed the signal, we know precisely which perturbations target something that is there and which target something that is not. That is exactly the knowledge real data cannot supply, and it is the minimum needed to measure a false-alarm rate at all. Section 5 addresses how far the resulting calibration transfers.

## 3 CONTROLS AND THEIR OPERATING CHARACTERISTICS

### 3.1 Definitions

A control is a map $\mathcal{P} : X \mapsto X'$ that ablates a hypothesised carrier. Band controls zero a spectral interval,

$$\mathcal{P}_{\text{band}}(X) = \mathcal{F}^{-1}\left\{\tilde{X}(f) \cdot \mathbb{1}[f \notin B]\right\}, \tag{3}$$

and spatial controls silence an electrode set, $\mathcal{P}_{\mathcal{R}}(X)_{c,t} = 0$ for $c \in \mathcal{R}$. A control *fires* when the induced accuracy drop exceeds a fixed threshold,

$$\Delta = a(X_{\text{test}}) - a(\mathcal{P}(X_{\text{test}})) > \tau, \qquad \tau = 0.10. \tag{4}$$

Over repeated datasets we estimate the two rates that define the instrument,

$$\text{sensitivity} = \Pr(\Delta > \tau \mid \text{target present}), \tag{5}$$

$$\text{false alarm} = \Pr(\Delta > \tau \mid \text{target absent}). \tag{6}$$

A control is trustworthy for a given decoder only when (5) is high *and* (6) is low *for that decoder.*

### 3.2 A minimal dissociation

Before the full grid, one controlled comparison isolates the mechanism. We plant the signal in motor-$\mu$ and build three decoders: Focused, a logistic model on two features (mean $\mu$ power over left- and right-motor channels); Broad, a logistic model on 38 features ($\mu$ and $\beta$ power at all 19 channels); and Tree, a random forest on the *identical* 38 features.

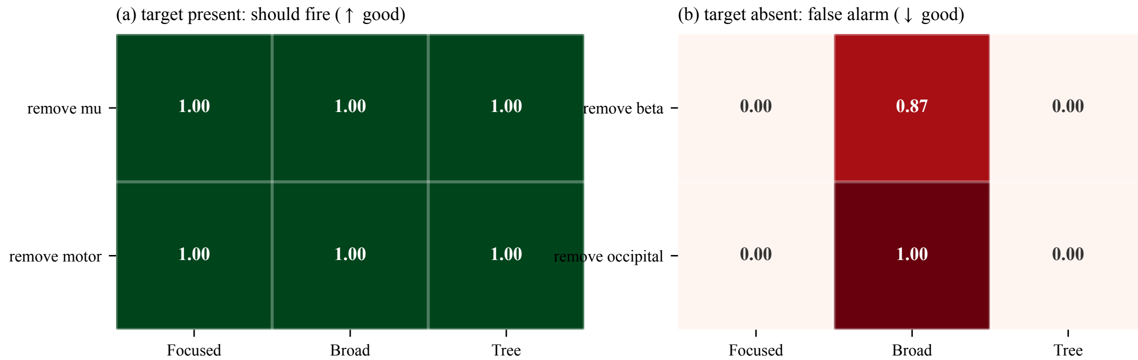


Fig. 2. Measured fire rates over 15 simulated datasets. (a) All three decoders detect the genuinely planted dependence. (b) The linear Broad decoder false-alarms on 87% of $\beta$ removals and 100% of occipital removals, where no signal exists. Tree sees the identical 38 features and never false-alarms.

Two controls target the planted signal (remove $\mu$, remove motor) and two target nothing (remove $\beta$, remove occipital). Results over 15 datasets appear in Figure 2.

All three decoders pass the sensitivity test: removing $\mu$ or silencing motor channels degrades every one of them, as it should. The false-alarm panel dissociates them completely. Broad fires on 86.7% of $\beta$ removals and 100% of occipital removals—perturbations that provably destroy no class information—while Focused and Tree never fire.

The explanation is mechanical and worth stating precisely. A linear model assigns nonzero weight to essentially every input, so zeroing *any* feature shifts its decision value and can push trials across the boundary; the drop reflects input perturbation, not dependence on the ablated carrier. A tree only splits on the features it uses, so zeroing an unused band or region leaves its decision path untouched. Since Broad and Tree consume identical inputs, the difference cannot be attributed to what the decoder can see. Same information, different model class, opposite reliability.

### 3.3 Measured operating characteristics at scale

The full study sweeps 50 ground-truth conditions (rhythm × region × strength × confound × two signal types), each with five network retrainings, on a 64-channel montage, across four decoder families: band-power with logistic regression, CSP with LDA [12], a random forest, and EEGNet [8]. A control is graded on a decoder only if that decoder cleared a decodability floor on the planted signal. Figure 3 reports the result.

Sensitivity is uninformative—every cell is 83–100%, so every control detects real dependence. The discrimination is entirely in the false-alarm panel. Averaged over probes, the false-alarm rates are 68.8% for band-power, 63.0% for the tree, 24.7% for CSP, and 1.3% for EEGNet. A "remove $\mu$" result that fires is thus near-conclusive for EEGNet and close to uninformative for band-power, on identical data with an identical perturbation.

The tree's position is instructive and refines the toy conclusion. At scale the random forest does *not* inherit the toy Tree's immunity: it halves band-power's occipital false alarms (33% versus 67%) but matches or exceeds it on band controls (75% on remove $\mu$, 83% on remove $\beta$). Selectivity in the splitting rule protects against spatial ablation, where irrelevant channels are simply never split on, but not against band ablation, which corrupts the very features the tree does use. The failure is therefore jointly attributable to the feature representation and the model class, and neither alone predicts reliability.

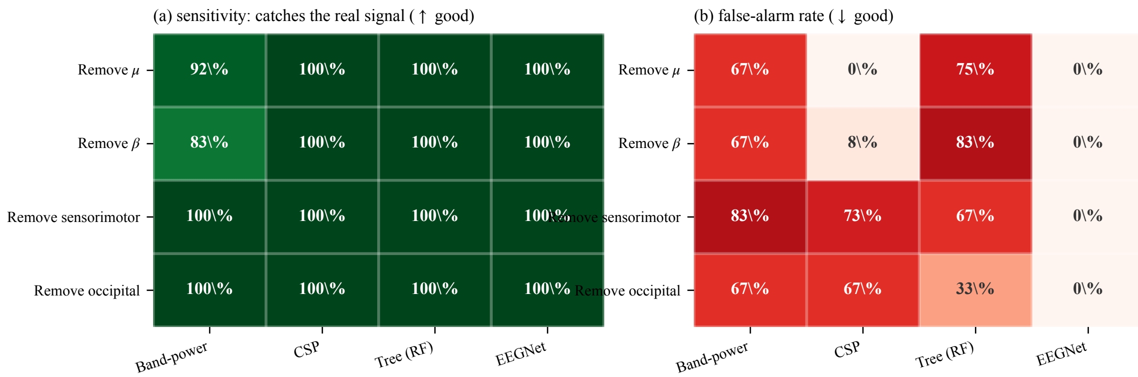


Fig. 3. Measured operating characteristics on the simulator grid. (a) Sensitivity is high everywhere: all four decoder families detect genuine dependence. (b) False-alarm rates differ by more than an order of magnitude. Band-power fires on 67–83% of perturbations whose targets are absent; EEGNet essentially never does.

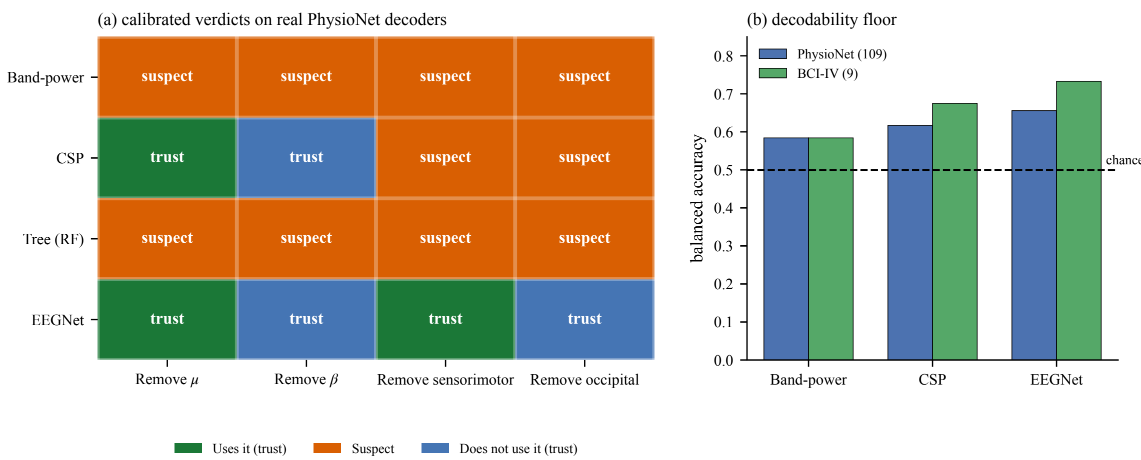


Fig. 4. (a) Calibrated verdicts on real PhysioNet decoders. Each cell attaches the measured reliability of Figure 3 to the observed accuracy drop, converting "the control fired" into a trust-weighted statement. Band-power and the tree earn no trustworthy verdict on any control; EEGNet earns four. (b) Held-out balanced accuracy: all decoders read motor imagery above chance.

## 4 AUDITING REAL DECODERS

We now apply the calibrated battery to decoders trained on PhysioNet EEGMMIDB [4, 15] with participant-held-out evaluation—every subject a decoder is tested on is one it never trained on. All three real decoders clear the decodability floor (Figure 4(b)); the BCI Competition IV-2a set [16] is excluded from the audit because nine participants render its test underpowered, which is a limitation rather than a result.

The verdicts are the paper's practical payoff. Every one of the four controls fires on band-power with drops of 0.03–0.08, and *none* of those firings is informative, because band-power's measured false-alarm rate on each is 67–83%. The same four controls applied to EEGNet yield drops of 0.01–0.09 that are fully interpretable: EEGNet depends on $\mu$ and on sensorimotor channels and does *not* depend on $\beta$ or on occipital channels—four decisive conclusions, because

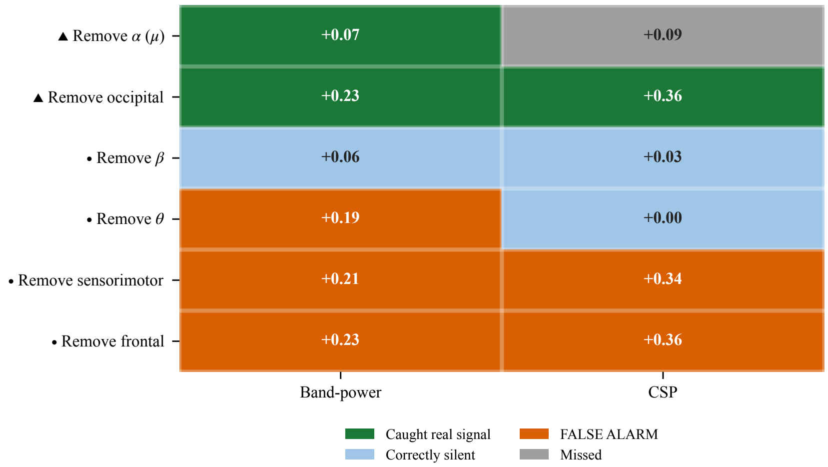


Fig. 5. Real-brain validation on eyes-open versus eyes-closed, 34 PhysioNet subjects, where physiology fixes the truth to occipital alpha. Triangles mark controls that should fire, dots those that should stay silent; cells give the accuracy drop. Band-power false-alarms on three of four absent targets, CSP on both spatial ones.

EEGNet's false-alarm rate is near zero. CSP is intermediate, trustworthy on the band controls and suspect on both spatial ones.

An uncalibrated report of the same experiment would have listed eight accuracy drops of comparable magnitude and drawn eight conclusions. Four of them would have been unfounded.

## 5 VALIDATION ON REAL BRAINS

The obvious objection is that all of the above is simulated. Real data cannot in general settle it, because measuring a false-alarm rate requires knowing that a target is absent, and on real brains that is unknown—which is why the simulator exists.

There is, however, a way around the impasse: choose a real task whose ground truth physiology already fixes. Eyes-closed versus eyes-open is the strongest such case in electroencephalography. Closing the eyes reliably increases occipital alpha, an effect known since the field's founding [3] and characterised in detail since [2]. On the resting recordings of the same PhysioNet participants, we therefore know the discriminative signal is occipital and in the alpha/$\mu$ band. Controls aimed at that signal *should* fire; controls aimed at $\beta$, $\theta$, sensorimotor, or frontal targets *should not.*

We decoded eyes-open versus eyes-closed on 34 subjects (band-power 73%, CSP 86% balanced accuracy—a real, strong signal) and graded the controls against that known truth. Figure 5 shows the outcome.

Band-power caught the genuine occipital-alpha dependence—and also fired on $\theta$ removal (+0.19), sensorimotor removal (+0.21), and frontal removal (+0.23), "detecting" three dependences that are not there. Its frontal false alarm is numerically as large as its true occipital detection (+0.23 each), so on this decoder the magnitude of a drop carries no information about whether the target was real. CSP false-alarmed on both spatial controls, with a frontal drop (+0.36) equal to its true occipital one.

This is the simulator-to-real bridge. The specificity failure predicted from planted ground truth appears, in the same form and on the same decoder families, on real human EEG in a task where the answer is independently known.

The caution is not a simulation artefact. We note the natural confound—volume conduction spreads occipital alpha to distant electrodes [9]—and observe that it strengthens rather than weakens the conclusion: a control that fires because of field spread is exactly a control that cannot localise the signal it claims to localise.

## 6 DISCUSSION

### 6.1 What the study establishes

(1) Perturbation controls have measurable operating characteristics, and those characteristics are *decoder-conditional*: false-alarm rates range from 1.3% to 68.8% across decoder families on identical data.
(2) Sensitivity is nearly uninformative in this setting; specificity does all the discriminating. Studies that report only that a control fired have reported the uninformative half.
(3) The failure is jointly a property of features and model class. A tree on identical features resists spatial ablation but not band ablation.
(4) Calibration changes real conclusions: the same battery on the same PhysioNet data yields four trustworthy verdicts for EEGNet and none for band-power.
(5) The specificity warning reproduces on real brains in a task with known physiology.

### 6.2 What it does not establish

Simulated ground truth is operational, not biological, so how completely the measured reliabilities transfer to every recording pipeline remains open—this is the study's honest crux, not a footnote. Each grid condition currently rests on a single simulated draw, so the reliability numbers carry no error bars; multi-draw replication is the obvious next step. The real-brain confirmation covers band and region controls on one alpha task, not the motor-imagery decoders themselves. Two older datasets, two-class motor imagery, and scalp EEG bound the scope, and no claim about cortical causality is made anywhere. The unit of inference must also stay explicit: trials estimate a subject-specific decoder, whereas subjects support population claims.

### 6.3 Recommendation

A perturbation result should not be reported as evidence unless the control's false-alarm rate has been measured for the decoder in question. Concretely, we recommend that papers using perturbation controls report, alongside each accuracy drop, the drop induced by at least one perturbation whose target is known to be absent for that decoder. That single addition converts an uninterpretable number into an interpretable one, and it is far cheaper than the calibration grid reported here.

## 7 CONCLUSION

Perturbation controls are instruments, and instruments require calibration. We measured the sensitivity and false-alarm rate of band and spatial controls against planted ground truth and found that the same control, on the same data, supports a decisive conclusion for one decoder and none at all for another—a factor of 50 in false-alarm rate between EEGNet and band-power. Applying the calibrated battery to real decoders reverses half the conclusions an uncalibrated reading would have drawn, and a real task with known physiology confirms the warning outside simulation. The deliverable is not a verdict on any particular decoder but a usable instrument: a way to say how much a perturbation result is worth, and the finding that the answer depends on what you pointed it at.

## REPRODUCIBILITY

The accompanying figure script regenerates the simulator panels and the full toy calibration live from the generator of Equations (1) and (2) using scikit-learn [10], and writes every measured operating characteristic, audit verdict, and eyes-open/closed drop to CSV. Simulation seeds are fixed, planted truth is defined before any control is evaluated, and catch and false-alarm regimes are reported separately for each decoder family.

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